

Electronic-Photonic Co-Optimization of Linear Drive Laser-Forwarded Coherent Silicon Photonic Transmitters for Co-Packaged Optical (CPO) Links

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Abstract—As data center performance scales rapidly with the highest-ever growth in network traffic, co-packaged optics (CPO) proves to be a potential solution for realizing high bandwidth density and energy-efficient interconnects within data centers. Coherent electro-optic transceivers in O-band are gaining much attention for scaling the bandwidth density of CPO interfaces. Co-design and co-optimization of these transceivers, while incorporating bandwidth limitations and link equalization capabilities, play a vital role in designing truly optimized energy-efficient systems. In most electro-optic transceivers, the overall power consumption is often dominated by the transmitter side, including the laser source. In this work, we first propose a modelling framework based on closed-form expressions for both the optical and electrical data path of a coherent transmitter. The modelling framework incorporates different link constraints such as data rate, input electrical swing, etc. Different technology constraints such as electrical and optical device parameters are also considered. Using this modelling framework, we propose a bandwidth-aware co-optimization scheme to optimize the overall energy-efficiency of the transmitter including the laser source. The co-optimization scheme assumes PAM4 electrical data and it leverages the coherent transmitter pulse-response to incorporate transmitter bandwidth limitations and host SerDes equalization capabilities. For 224 Gb/s QAM-16 operation, an optimized unequalized linear-drive transmitter is projected to reach an energy-efficiency of 2 pJ/bit including the laser power consumption. Energy-efficiency can be reduced down to 1.4 pJ/bit with a 1-tap TX feed-forward equalization (FFE) and to 1.2 pJ/bit with a 1-tap decision-feedback equalization (DFE) in the host SerDes.

Index Terms—Coherent, co-optimization, co-packaged optics (CPO), modulator, silicon photonics.

I. INTRODUCTION

WITH the rapid advancement and adoption of artificial intelligence (AI) in possibly every sector of today's world, the largest-ever volume of data is being managed by

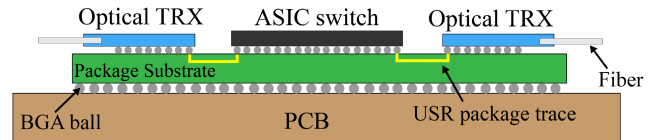


Fig. 1. Monolithic optical transceiver chiplets co-packaged with ASIC switch.

today's data centers. This gives rise to a massive demand for increasing the bandwidth density and energy-efficiency of data communication chips sitting at different levels hierarchy within the data center network. Even as optical interconnects continue replacing the electrical interconnects due to its limited bandwidth-reach product [1], almost every switching operation is still being performed in the electrical domain, driving the need for efficient E/O and O/E interfaces. In recent years, with the evolution of co-packaged optics (CPO) [2], [3], [4], [5], there has been a significant push for moving optics closer to the switch ASIC as shown in Fig. 1. Co-packaging optics with switch ASIC obviates the need for power-hungry DSP-based retimers traditionally used in pluggable optical modules, thus lowering system power significantly. Moreover, CPO enables much higher package escape bandwidth by means of dense electrical connection between switch ASIC and optical transceiver. However, to keep data center performance scaling uninterrupted in future, CPO bandwidth density needs to be scaled continuously. In addition, low power consumption should be maintained to avoid thermal challenges due to co-packaging. At the same time, CPO should maintain compatibility with the existing Ethernet standards such as DR, FR4 etc. As scaling electrical data rate is becoming significantly challenging, there is a growing need for exploring other signaling dimensions for scaling CPO bandwidth density. At the same time, it is also important to look for optimized energy-efficient optical transceivers.

Coherent optical communication poses to be one of the potential solutions for scaling bandwidth density due to its high spectral efficiency. Contrary to traditional intensity-modulation direct detection (IMDD) based approach, coherent modulation utilizes additional degrees of freedom such as phase and polarization properties of light to increase channel spectral efficiency. Although coherent optical links are widely adopted for long-haul communication in C-band [6], they have not been deployed for intra-data-center links so far mainly due to the

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power-hungry digital signal processing (DSP) which deals with carrier recovery, timing recovery and different optical fiber non-idealities such as chromatic dispersion (CD), polarization-mode dispersion (PMD). However, as [7] analytically showed, the DSP of a coherent transceiver can be simplified to a large extent for short-reach (< 2 km) intra-data center applications in O-band. For example, [8] experimentally demonstrated a DSP-free 400Gb/s dual-polarization (DP) 16-QAM intra-data center transceiver using discrete electrical and optical components. Recently, [9] presented an integrated 224Gb/s DP-QPSK link operating in O-band. Although these solutions address the challenge of scaling bandwidth density, nonetheless, they lack co-design and co-optimization of electronics and photonics leading to less energy-efficient systems.

Overall power consumption of the optical transceiver is often dominated by the transmitter side which includes the laser power [8], [10]. Coherent optical transceivers typically utilize Mach-Zehnder Modulators (MZMs). An MZM driver typically consumes high power due to high voltage-swing requirement and large load-capacitance of MZM. Moreover, linear-drive CPO applications [4] utilize linear MZM drivers, which consume more power for maintaining linearity as compared to the switching drivers for similar voltage swing. Similarly, the required laser power is also significantly high due to optical data path loss which includes the insertion loss of all optical components and modulation loss of the MZM. A laser-forwarded scheme [11] can lower the required laser power by averaging the path losses. Moreover, it obviates the need for any optical PLL in the coherent receiver, simplifying the receiver design. Nonetheless, several electro-optic design trade-offs, such as driver and laser powers, motivates co-optimizing the transmitter. In this work, we investigate the electrical and optical data path of an MZM-based laser forwarded QAM-16 coherent transceiver for linear-drive CPO application. The proposed modeling approach combines the behavioral model of the optical data path with that of the electrical data path to obtain a closed-form expression for the electro-optic response of the coherent transmitter. Similarly, to estimate the required transmitter and laser power consumption, a closed form expression for the receiver sensitivity has been derived through modeling the coherent receiver data path. These closed-form expressions dictate the dependence of the link performance on different optical and electrical design parameters, technology constraints and link constraints. Unlike previously reported electronic-photonics modeling approaches [12], [13], the proposed approach captures the electrical bandwidth limitations of the driver as well as the electro-optic modulator, which is significantly crucial for symbol-rate as high as 56 GSym/s. Utilizing this behavioral electro-optic model of the coherent transceiver, we then present an electronic-photonics co-optimization framework based on the transmitter electro-optic pulse-response. Several electrical and optical design parameters such as driver parameters, MZM length, laser power splitting ratio etc. are optimized to improve the energy-efficiency of the coherent transmitter including the laser. The proposed co-optimization framework is also capable of incorporating host SerDes equalization settings, which has been demonstrated by a 1-tap DFE and 1-tap TX-FFE example cases.

The rest of the paper is organized as follows: Section II provides a high-level overview of the laser-forwarded coherent transceiver architecture analyzed in this work. Section III models the optical data path of the coherent transceiver, followed by the MZM driver modelling in Section IV. Then, using the results from prior sections, we propose in Section V a bandwidth-aware transmitter electro-optic co-optimization framework based on peak-distortion analysis of electro-optic pulse-response. Finally, Section VI concludes the paper.

II. MZM-BASED LASER-FORWARDED COHERENT OPTICAL TRANSCEIVER

The coherent optical transceiver architecture considered in this work is shown in Fig. 2. The ASIC switch is surrounded by several optical transceivers (OTRX) which are mounted on the same package substrate as the ASIC. The electrical interconnects between the ASIC and the OTRX are realized by low-loss ultra-short reach (USR) package traces. These USR channels are typically < 10 mm long and the insertion loss is lower than 2 dB at 28 GHz [14]. The electronic and photonic components on the OTRX are considered to be monolithically integrated into a single chip as in [11], [15]. This monolithic integration significantly improves performance and power consumption by eliminating die-to-die interface parasitic between electronic and photonic components and allowing dense integration of circuits and photonics [16].

A. Transceiver Architecture

In the system under consideration, the external laser is split into two portions: one portion goes to the transmitter and is modulated by the electro-optic IQ modulator, whereas another portion is forwarded to the coherent receiver as a local oscillator (LO). The fraction of total laser power that is forwarded to the receiver is denoted as k . On the transmitter side, the input laser enters the chip through an edge coupler and is split equally between the I and Q paths. The 50:50-splitter can be realized with a 1:2-MMI or a 3-dB directional coupler. In each of the I and Q paths, the light gets modulated through the MZM by the incoming electrical data. Among different realizations of the MZM in a silicon-photonics process, the most common one utilizes depletion-mode PN-junction phase shifters on its arms [17]. Although this work assumes a segmented MZM architecture, the analysis can be readily extended to travelling wave MZMs [17]. The MZMs are driven by linear high-swing electrical drivers which receive PAM4 data-streams from the SerDes TX in the host ASIC. Although, this work assumes an electrical input data-rate of 112 Gb/s, the analyses and results can be readily extended to > 200 Gb/s electrical data-rates. The modulated light then goes through a thermal phase-shifter which creates a 90° phase-shift between the I and Q paths. As both the I- and Q- modulated signals have 4-levels, the IQ combined light becomes QAM-16, which is then transmitted to some other CPO module within the data center.

The coherent RX receives the transmitted optical signal along with the forwarded LO and combines them in a 90° -optical

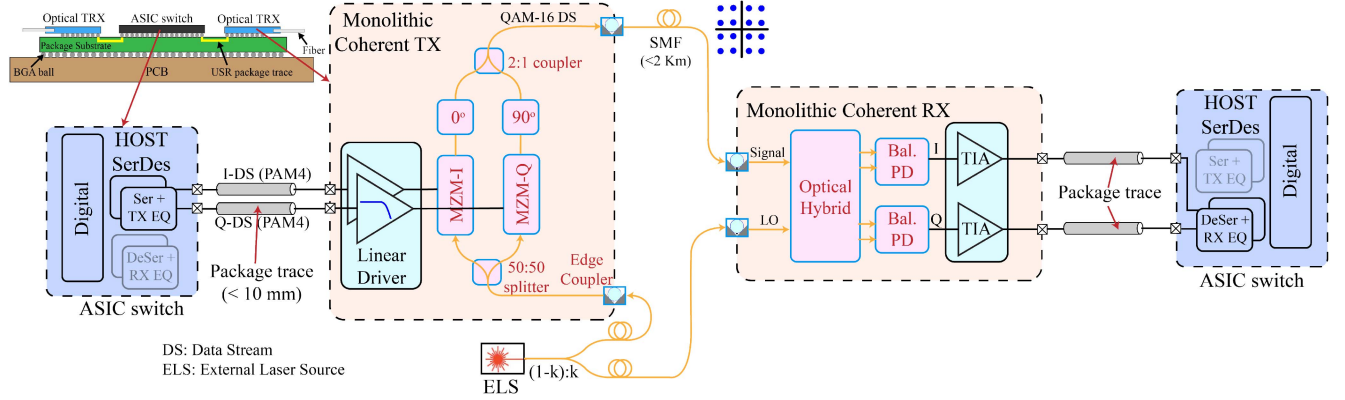


Fig. 2. Simplified block diagram of a laser-forwarded coherent optical transceiver (TRX) co-packaged with an ASIC switch. The electrical path from the ASIC switch to the optical TRX is realized by an USR interconnect having a length lower than 10 mm and insertion loss smaller than 2 dB at 28 GHz [14]. The optical TRX monolithically integrates the electrical components such as the driver, TIA, and the optical components.

hybrid. In practice, more specifically for dual-polarization systems, before the optical hybrid, both the signal and LO pass through polarization controllers which perform necessary polarization rotations to the signal and LO [8]. The optical hybrid separates the I and Q data and sends them to a pair of balanced photo-detectors which generate the I and Q signals respectively. These I and Q current signals are converted to voltages using trans-impedance amplifier (TIA) stages and then sent to the corresponding host SerDes for further processing and/or equalization.

B. Scope of Co-Optimization

The overall link energy efficiency is governed by the power consumption of the host SerDes and the optical TRX. Both of these are often designed as separate IPs and the host SerDes is typically optimized for the specific type of electrical interconnect (e.g. USR, XSR, LR etc.). However, the optical TRX can be designed with the awareness of the signal strength and equalization capabilities of the host SerDes. This work focuses on the co-optimization of the photonic and electronic design parameters of the coherent optical TRX, while optimizing the coherent transmitter energy efficiency including the laser. However, estimating the energy efficiency of the host SerDes, given some specific equalizer or signal swing settings, is beyond the purview of this work.

In the design of the electro-optic data path of the coherent transceiver, several design choices are involved which govern the energy efficiency of the link. For example, to increase the transmitted optical swing or optical modulation amplitude (OMA), one can increase the driver voltage gain, increase the MZM length, or increase the laser power. The increase in optical phase shift obtained by increasing driver voltage gain directly trades with the transmitter linearity and bandwidth. Longer MZM creates larger phase shift, while at the same time, increases MZM insertion loss and also degrades the electro-optic bandwidth of the transmitter by offering larger load capacitance. Similarly, increasing the laser power increases the shot-noise of the balanced PDs, thus degrading coherent RX sensitivity.

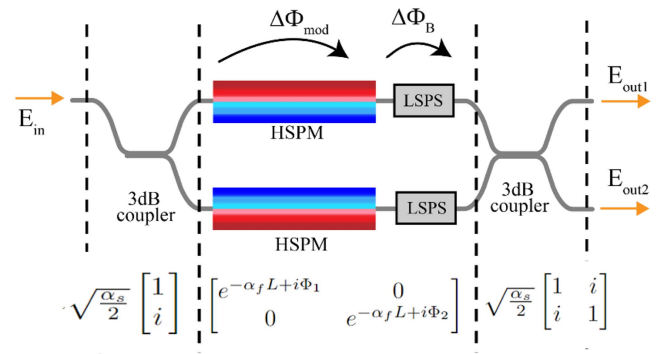


Fig. 3. MZM diagram with transfer matrices of difference sub-components. HSPM and LSPS represents the high-speed phase modulator and low-speed phase shifter respectively.

Moreover, photo diodes have certain power handling capability which limits the maximum LO laser power. The laser-splitting ratio is another design choice which affects the optical signal swing and RX sensitivity. Hence, each of the design choices has specific implications on the total energy-efficiency of the transmitter, making it important to co-optimize the link in terms of all the design parameters. In the next section, we model the entire optical data-path of the coherent transceiver.

III. OPTICAL DATA PATH

A. TX Optical Data Path

1) *MZM Electro-Optic Model:* The fundamental component of the optical path of the transmitter described in the previous section is the electro-optic MZM, which works on the principle of controlling the interference of light according to the applied voltage. As shown in Fig. 3, the incoming light gets split into the two arms of the MZM. The high-speed electro-optic phase-modulators (HSPM) on both arms provide phase shifts to the propagating light depending on the applied voltages. The physical phenomenon which accounts for the voltage dependent phase-shift is known as the free-carrier plasma-dispersion

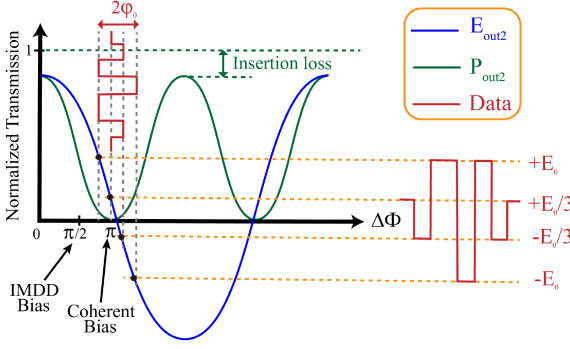


Fig. 4. Normalized field and power transmission of the MZM with the phase-shift difference between arms. For a PAM4 electrical data-stream, the corresponding phase-shift and optical field shows similar PAM4 behavior.

effect [18]. Finally, after getting combined, the light shows modulated behaviour in both amplitude and phase depending on the phase shift difference between the two MZM arms. From a modelling perspective, each sub-block of the MZM can be represented through a transfer matrix which relates the input and output optical fields of the propagating light. Fig. 3 shows these transfer matrices for the input and output 3dB-splitters or directional couplers and the two MZM arms. Here, α_s is the power insertion-loss of the 3dB-splitter, α_f is the field absorption coefficient of the HSPM, L is the length of the HSPM and Φ_1, Φ_2 are the phase shifts of the two MZM arms respectively. The loss numbers depend on the process and device design, and are often determined by optical simulations or device measurement results. The insertion loss of the low-speed phase-shifter used for biasing the MZM is often very small compared to other losses and hence is neglected in this analysis. Note that a 2×2 -directional coupler is used at the output as one of the outputs can be used to monitor the light and set proper biasing of the MZM. The output field expression can be found by cascading the transfer matrices as shown below,

$$\begin{bmatrix} E_{out1} \\ E_{out2} \end{bmatrix} = \frac{\alpha_s}{2} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} \begin{bmatrix} e^{-\alpha_f L + i\Phi_1} & 0 \\ 0 & e^{-\alpha_f L + i\Phi_2} \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} E_{in} \quad (1)$$

This gives the output optical fields to be

$$E_{out1} = E_{in} \alpha_s e^{-\alpha_f L} e^{i(\Phi_{avg} + \pi/2)} \sin\left(\frac{\Delta\Phi}{2}\right) \quad (2a)$$

$$E_{out2} = E_{in} \alpha_s e^{-\alpha_f L} e^{i(\Phi_{avg} + \pi/2)} \cos\left(\frac{\Delta\Phi}{2}\right) \quad (2b)$$

where $\Phi_{avg} = (\Phi_1 + \Phi_2)/2$ is the mean phase shift and $\Delta\Phi = (\Phi_1 - \Phi_2)$ is the phase-shift difference between the two arms of the MZM. Moreover, $\Delta\Phi$ can be divided into the static phase-shift term, ϕ_B and the modulation term, ϕ_{mod} . As shown in Fig. 4, for IMDD applications, the MZM is quadrature biased, i.e., $\phi_B = \pi/2$ and for coherent applications, it is null-biased, i.e., $\phi_B = \pi$. The phase-shift of an electro-optic phase-shifter is related to the applied voltage through the modulation efficiency, $V_\pi L_\pi$, where V_π denotes the amount of voltage needed to create

a π phase-shift to a phase-shifter with length L_π . This $V_\pi L_\pi$ product serves as the figure-of-merit for an electro-optic phase shifter. It is dependent on the type of the phase shifter (e.g. depletion-based, carrier-injection-based etc.), phase-shifter geometry and design. The modulation efficiency typically degrades after a certain frequency of the applied electrical signal governing the intrinsic electro-optic (EO) bandwidth of the MZM. Considering a differential push-pull MZM architecture [17] and a differential peak swing of $V_{dr,p}$, the DC modulation phase-shift can be given as

$$\phi_{mod,DC} = \pi \times \frac{V_{dr,p} L}{V_\pi L_\pi} \quad (3)$$

The modulation phase-shift drops at higher frequencies due to the limited electro-optic (EO) bandwidth of the phase-shifter. Assuming a first order roll-off, the frequency dependent modulation phase-shift can be given as

$$\phi_{mod}(s) = V_{dr}(s) \times H_{ps}(s) \quad (4a)$$

$$= V_{dr}(s) \times \frac{H_{ps,0}}{1 + s/\omega_{EO}} \quad (4b)$$

where $V_{dr}(s)$ is the drive voltage in frequency domain, $H_{ps}(s)$ is the phase shifter transfer function and ω_{EO} is the intrinsic EO bandwidth of the MZM. The DC value of $H_{ps}(s)$ can be obtained using (3) as $H_{ps,0} = \pi L/V_\pi L_\pi$.

In (2a,2b), $\Delta\Phi_{avg}$ remains constant for differential push-pull operation. Hence, ignoring the static phase shift of the output light, the output fields for coherent application can be written as

$$E_{out1} = E_{in} \alpha_s e^{-\alpha_f L} \cos\left(\frac{\phi_{mod}}{2}\right) \quad (5)$$

$$E_{out2} = -E_{in} \alpha_s e^{-\alpha_f L} \sin\left(\frac{\phi_{mod}}{2}\right) \quad (6)$$

Here, E_{out2} is the signal and E_{out1} can be sent to an on-chip optical power monitor to set the proper MZM bias (i.e., quadrature vs null). Fig. 4 visually shows the PAM4 levels for modulation phase-shift and output optical field corresponding to an electrical PAM4 data. Note that, although phase shift, ϕ_{mod} is linearly proportional to the electrical swing ((4)), the optical field has a sinusoidal relationship with the phase shift (6). This results in linearity degradation of optical field for a very large electrical swing. To quantify the degradation in optical field linearity, Fig. 5 shows the change in ratio of level mismatch, R_{LM} [20], a standard PAM4 linearity metric, with input electrical swing. This analysis considers photonic device parameter values as $V_\pi L_\pi = 16$ Vmm and $\alpha_f = 4$ dB/cm. As it can be seen, a longer MZM and very high value of electrical swing start degrading optical field linearity. As an example, for a 1mm long MZM and with a 98% R_{LM} target, the MZM can sustain a 7.5V peak-peak differential input electrical swing before running into linearity penalty. The maximum achievable voltage swing offered by linear drivers in advanced CMOS processes [21] is typically much smaller than this. Therefore, the overall linearity of the transmitter often gets determined by the electrical driver rather than the MZM. Fig. 5 also depicts the variation of MZM modulation loss which is the ratio of the

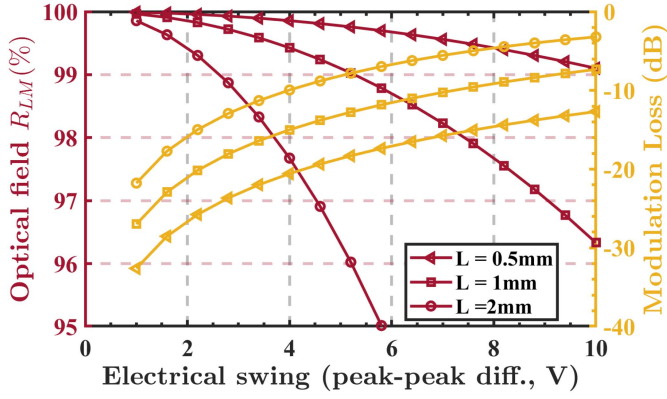


Fig. 5. Variation of optical field linearity (in terms of PAM4 R_{LM}) and modulation loss of an MZM with input electrical swing. Shown for 3 different MZM lengths, L .

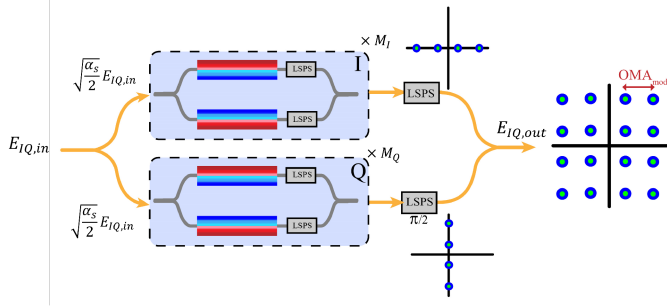


Fig. 6. IQ electro-optic modulator utilizing MZMs. LSPS in the I- and Q-path is used for creating the $\pi/2$ phase-shift between the I and Q paths.

optical signal power and the maximum achievable optical power out of the MZM. A higher electrical swing and longer MZM helps reducing the modulation loss of the MZM.

2) **MZM Based IQ Modulator:** Two MZMs can be placed in the I and Q path to form an IQ-modulator as shown in Fig. 6. Two PAM4 electrical data streams modulate each MZM, resulting in a QAM-16 optical data, where each symbol carries 4-bits. Taking into account the insertion loss for the input splitter and output combiner, transfer matrix based approach can be used to find out the output field expression as discussed previously. For an input field $E_{IQ,in}$ and splitter/combiner loss α_s the output field can be derived as

$$E_{IQ,out} = \alpha_s \frac{E_{IQ,in}}{\sqrt{2}} M_{IQ} e^{i\phi_{IQ}} \quad (7)$$

where,

$$M_{IQ} e^{i\phi_{IQ}} = M_I + iM_Q \quad (8a)$$

$$= \alpha_s e^{-\alpha_f L} \left[\sin\left(\frac{\phi_{mod,I}}{2}\right) + i \sin\left(\frac{\phi_{mod,Q}}{2}\right) \right] \quad (8b)$$

Note that, here, $M_{IQ} = |M_I + iM_Q|$ and $\phi_{IQ} = \arctan(M_I + iM_Q)$. The modulation phase-shift terms $\phi_{mod,I}$ and $\phi_{mod,Q}$ are related to the electrical drive signal as

described in the previous sub-section,

$$\phi_{mod,I}(s) = V_{dr,I}(s) \times H_{ps,I}(s) \quad (9)$$

$$\phi_{mod,Q}(s) = V_{dr,Q}(s) \times H_{ps,Q}(s) \quad (10)$$

An additional thermo-optic phase-shifter is used in the Q-path to create a DC phase-shift difference of $\pi/2$ between the I and the Q paths. Note that M_I and M_Q are the normalized transfer functions of the I and Q MZM respectively. For intensity modulation, the transmit optical modulation amplitude (OMA) is typically given by the difference between the zero and one optical power levels. In a similar way, for coherent modulation, the OMA of the IQ-modulator can be defined as the optical power distance between the adjacent constellation points in the IQ-plane. In presence of noise and inter-symbol interference (ISI), the mean locations of the constellation points should be considered for modulator OMA. To make the modulator OMA independent of the laser power, it is often normalized by the input optical power [12]. Considering the fact that optical power is the square of the field and taking into account the phase modulation levels discussed earlier, the expression for the normalized OMA of the IQ-modulator, OMA_{mod} can be derived as,

$$OMA_{mod} = \frac{\text{mean distance of adjacent constellation points}}{\text{input power}} \quad (11a)$$

$$= \frac{\left[\frac{2}{3} \frac{E_{IQ,in}}{\sqrt{2}} \alpha_s^2 e^{-\alpha_f L} \sin\left(\frac{\phi_0}{2}\right) \right]^2}{E_{IQ,in}^2} \quad (11b)$$

$$= \frac{2}{9} e^{-\alpha_p L} \alpha_s^4 \sin^2\left(\frac{\phi_0}{2}\right) \quad (11c)$$

$$= \frac{\alpha_s^2}{9} M_{IQ,max}^2 \quad (11d)$$

where $\alpha_p = 2\alpha_f$ is the power absorption coefficient of the phase shifter. Here, we assume the peak differential modulation phase-shift in both the I and Q paths is ϕ_0 . Note that, in this analysis $E_{IQ,in}$ is the on-chip optical field that is coupled from the external laser source. Meaning, $E_{IQ,in} = \sqrt{(1-k)\alpha_c} E_{laser}$, where k is the splitting ratio of the laser-forwarding scheme that denotes the fraction of laser power sent to the receiver as LO and α_c is the power insertion loss of the edge couplers used to couple light into and out of the chip. In the same way, using (7) the optical field in the fiber which comes out of the chip can be expressed in terms of the laser optical field, E_{laser} as

$$E_{TX} = \sqrt{\alpha_c} E_{IQ,out} \quad (12a)$$

$$= \sqrt{(1-k)\alpha_c} \alpha_s \frac{E_{laser}}{\sqrt{2}} M_{IQ} e^{i\phi_{IQ}} \quad (12b)$$

B. RX Optical Data-Path

To derive the expression for the required laser power for meeting a certain bit error rate (BER) specification, it is important to understand and model the RX optical sensitivity. The two major sources of noise in an optical receiver are the shot noise of the photodiodes and thermal noise of the electrical front-end

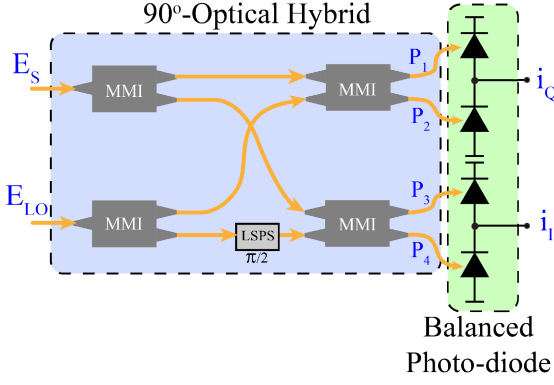


Fig. 7. Demodulation of IQ-data at receiver. In place of Multi-Mode Interferometers (MMIs), 3-dB couplers can also be used for splitting/combining light.

including the TIA. For an IMDD link, as shot noise of the photodetector is comparatively smaller due to the absence of LO, the sensitivity is mostly governed by the latter. However, for coherent links, due to the large LO power, shot noise can be significant compared to the TIA thermal noise [11]. Moreover, for a monolithic process, TIA thermal noise can be even smaller due to absence of pad capacitance which relaxes the feedback resistor value of the TIA reducing the noise floor [15].

1) *IQ Demodulation*: As discussed previously, in a laser-forwarded architecture, the transmitted IQ-modulated optical signal is combined with the forwarded LO at the RX through a 90°-hybrid and sent to a pair of balanced photodiodes to demodulate the input optical signal into I and Q data as shown in Fig. 7. In practice, an additional phase tuner is used in the LO path to de-rotate the IQ constellation relative to LO phase and remove any IQ-crosstalk. Similar transfer matrix based approach can be used to derive the output field expressions of the 90°-hybrid and thereby the output of the balanced PDs.

The signal and LO fields, E_s and E_{LO} , which are the inputs to the 90°-hybrid, can be given by

$$E_s = \sqrt{\alpha_c} E_{TX} \quad (13a)$$

$$= \alpha_c^{\frac{3}{2}} \alpha_s \sqrt{\frac{1-k}{2}} E_{laser} M_{IQ} e^{i\phi_{IQ}} \quad (13b)$$

$$= E'_s M_{IQ} e^{i\phi_{IQ}} \quad (13c)$$

$$E_{LO} = \sqrt{\alpha_c \alpha_r k} E_{laser} e^{i\phi_{LO}} \quad (14a)$$

$$= E'_{LO} e^{i\phi_{LO}} \quad (14b)$$

where α_r is the loss of LO phase tuner and ϕ_{LO} is the static phase offset between the signal and LO. Now the output fields of the optical hybrid can be expressed as,

$$E_{H1} = \frac{1}{2} \sqrt{\alpha_h} (E_s + iE_{LO}) \quad (15a)$$

$$E_{H2} = \frac{1}{2} \sqrt{\alpha_h} (iE_s + E_{LO}) \quad (15b)$$

$$E_{H3} = \frac{1}{2} \sqrt{\alpha_h} (E_s + E_{LO}) \quad (15c)$$

$$E_{H4} = \frac{i}{2} \sqrt{\alpha_h} (E_s - E_{LO}) \quad (15d)$$

where α_h is the optical insertion loss of the 90°-hybrid. To derive the output current of the balanced detector, these field expressions need to be converted to corresponding optical power which is the magnitude square of the field. Thus, the corresponding powers can be derived as,

$$P_1 = \frac{\alpha_h}{4} [E_s'^2 M_{IQ}^2 + E_{LO}'^2 + 2E_s' E_{LO}' M_{IQ} \sin(\phi_{IQ} - \phi_{LO})] \quad (16a)$$

$$P_2 = \frac{\alpha_h}{4} [E_s'^2 M_{IQ}^2 + E_{LO}'^2 - 2E_s' E_{LO}' M_{IQ} \sin(\phi_{IQ} - \phi_{LO})] \quad (16b)$$

$$P_3 = \frac{\alpha_h}{4} [E_s'^2 M_{IQ}^2 + E_{LO}'^2 + 2E_s' E_{LO}' M_{IQ} \cos(\phi_{IQ} - \phi_{LO})] \quad (16c)$$

$$P_4 = \frac{\alpha_h}{4} [E_s'^2 M_{IQ}^2 + E_{LO}'^2 - 2E_s' E_{LO}' M_{IQ} \cos(\phi_{IQ} - \phi_{LO})] \quad (16d)$$

Considering the responsivity $\mathcal{R}$ of the photo detector, the received I and Q currents can be given as,

$$i_{RX,I} = \mathcal{R}(P_3 - P_4) \quad (17a)$$

$$= \mathcal{R} \alpha_h E_s' E_{LO}' M_{IQ} \cos(\phi_{IQ} - \phi_{LO}) \quad (17b)$$

$$= \mathcal{R} \alpha_h \alpha_c^2 \alpha_s \sqrt{\frac{\alpha_r k(1-k)}{2}} P_{laser} M_{IQ} \cos(\phi_{IQ} - \phi_{LO}) \quad (17c)$$

$$i_{RX,Q} = \mathcal{R} \alpha_h \alpha_c^2 \alpha_s \sqrt{\frac{\alpha_r k(1-k)}{2}} P_{laser} M_{IQ} \sin(\phi_{IQ} - \phi_{LO}) \quad (18)$$

Here, $P_{laser} = |E_{laser}|^2$ is the total laser power. Eqs. ((17), (18)) indicate that if $\phi_{LO} = 0$, the received currents are directly proportional to the transmitted I and Q signals. Hence, for proper demodulation, the phase offset of the LO should be zero. This work assumes a standard LO recovery scheme to achieve IQ demodulation such as [22] as an example. With this zero phase offset assumption, the received currents can be expressed as,

$$i_{RX,I} = \mathcal{R} \alpha_h \alpha_c^2 \alpha_s \sqrt{\frac{\alpha_r k(1-k)}{2}} P_{laser} M_I \quad (19)$$

$$i_{RX,Q} = \mathcal{R} \alpha_h \alpha_c^2 \alpha_s \sqrt{\frac{\alpha_r k(1-k)}{2}} P_{laser} M_Q \quad (20)$$

2) *RX Sensitivity and Laser Power Splitting Ratio*: As the signal swing in both the I and Q paths are ideally the same, consider only one of the received currents and analyze the optical sensitivity. Note that the received signal current has 4 amplitude levels and the peak-to-peak value can be given by,

$$i_{sig,pp} = 2\mathcal{R} \alpha_h \alpha_c^2 \alpha_s \sqrt{\frac{\alpha_r k(1-k)}{2}} P_{laser} \left[\alpha_s e^{-\alpha_f L} \sin\left(\frac{\phi_0}{2}\right) \right] \quad (21a)$$

$$= 3\mathcal{R} \alpha_h \alpha_c^2 \sqrt{\alpha_r k(1-k)} P_{laser} \sqrt{OMA_{mod}} \quad (21b)$$

Equation (11c) has been used to get the relation of received current amplitude with the normalized OMA of the IQ-modulator in transmitter chip. From (21b) one can find out the laser power required to obtain some desired signal swing at the receiver as,

$$P_{laser} = \frac{i_{sig,pp}^2}{3R\alpha_h\alpha_c^2\sqrt{\alpha_rk(1-k)OMAMod}} \quad (22)$$

This is the total laser power drawn from the external laser source (ELS) to both the transmitter and receiver. The ELS wall-plug power P_{ELS} can be found by dividing it with the wall-plug efficiency, η_w [12], i.e. $P_{ELS} = P_{laser}/\eta_w$. The optical energy efficiency, $E_{b,laser}$ is defined as the energy spent for the laser to transmit and receive a single bit of data and its derived as below,

$$E_{b,laser} = \frac{P_{laser}}{\eta_w\eta_m(4f_{symbol})} \quad (23a)$$

$$= \frac{i_{sig,pp}^2}{12R\alpha_h\alpha_c^2\sqrt{\alpha_rk(1-k)OMAMod}\eta_w\eta_m f_{symbol}} \quad (23b)$$

where η_m is the link margin [12] to take into account additional sources of optical loss such as loss due to optical fiber and waveguides and f_{symbol} is the symbol rate of the transmitted data. The factor of 4 accounts for the fact that 4-bits are transmitted in a symbol period of $1/f_{symbol}$ as it utilizes QAM16 modulation.

Equation (21b) indicates that for a fixed laser power and transmitter $OMAMod$, the received signal swing is maximum when the laser splitting-ratio, $k = 0.5$. However, the PD shot-noise is also proportional to the splitting ratio as will be shown. The optimum splitting-ratio is the one which maximizes the signal-to-noise ratio (SNR) instead of solely maximizing the signal amplitude. Note that the total noise at the input of the TIA is the sum of photo-detector shot noise and TIA thermal noise. TIA thermal noise, $\bar{i}_{nT}$ depends on the RX-frontend design and the PD shot noise can be derived as below,

$$\bar{i}_{ns}^2 = 2qR(P_3 + P_4)\Delta f_{ns} \quad (24a)$$

$$= 2qR \left[\frac{\alpha_h}{2} (E_s'^2 M_{IQ}^2 + E_{LO}^2) \right] \Delta f_{ns} \quad (24b)$$

$$= qR\alpha_h \left(\alpha_c^3 \alpha_s^2 \frac{(1-k)}{2} M_{IQ}^2 + \alpha_c \alpha_r k \right) P_{laser} \Delta f_{ns} \quad (24c)$$

here, Δf_{ns} is the equivalent shot-noise bandwidth. From the constellation of QAM-16 signal we can understand that M_{IQ}^2 can have 3 different values. Hence, we can consider the maximum value for worst case noise estimation. Hence, using (11d), the expression for shot-noise can be found as

$$\bar{i}_{ns}^2 = qR\alpha_h \left(\frac{9}{2} \alpha_c^3 (1-k) OMAMod + \alpha_c \alpha_r k \right) P_{laser} \Delta f_{ns} \quad (25)$$

The first term in (25) corresponds to the shot noise contributed by the signal and the second term, which is typically orders of magnitude greater than the first term, corresponds to the shot-noise contributed by the LO. Note that the rms value of shot noise increases with square-root of the laser power, whereas the signal

current amplitude increases linearly with the laser power ((21b)). Hence, the SNR due to only shot-noise increases linearly with increasing laser power. The bit-error-rate (BER) at the input of the TIA can be given by [23],

$$BER = \frac{3}{8} \operatorname{erfc} \left(\frac{\text{inner eye-height}}{2\sqrt{2} \times \text{total RMS noise}} \right) \quad (26a)$$

$$= \frac{3}{8} \operatorname{erfc} \left(\frac{i_{sig,pp}}{6\sqrt{2} \times \sqrt{\bar{i}_{ns}^2 + \bar{i}_{nT}^2}} \right) \quad (26b)$$

Equation (26b) captures the dependence of BER over different design parameters and indicates the need to optimize them for minimizing the BER. For example, both the received signal, $i_{sig,pp}$ ((21b)) and shot noise, $\bar{i}_{ns}$ ((25)) are dependent on the laser splitting ratio, k . Hence, there exists an optimal k which minimizes the BER. Similarly, $i_{sig,pp}$ is linearly proportional to the laser power, P_{laser} , whereas, $\sqrt{\bar{i}_{ns}^2 + \bar{i}_{nT}^2}$ has a square-root dependence on P_{laser} indicating improvement of BER on increasing P_{laser} . However, note that (26b) assumes that the inner-eye height is one-third of the peak-to-peak signal, $i_{sig,pp}$ which is only valid when the transmitter does not have any bandwidth limitation. However, for a very high symbol-rate as 56 GSym/s, the finite electrical bandwidth of the MZM driver and the finite EO bandwidth of the MZM cause settling time issues and create significant ISI reducing the inner eye-height. This will be analyzed in greater detail in Section V where we propose a bandwidth-aware co-optimization framework.

IV. LINEAR MZM DRIVER

The key requirements for the MZM driver in a linear-drive application are linearity, high-swing, high bandwidth and higher power efficiency. Linear high-swing drivers in literature are common in Silicon-Germanium (SiGe) BiCMOS processes [6], [25]. However, as compared to SiGe, standard CMOS process offers higher power efficiency, lower cost, higher yield and possibility of integration with existing CMOS solutions. Although recently some high-swing drivers were being reported in the CMOS and SOI processes [21], [26], [27], [28], most of them produce rail-to-rail output and hence, are suitable only for NRZ type of data. For PAM4 applications, these switching drivers are segmented into MSB and LSB segments and the data-streams are either electrically or optically combined to produce PAM4 modulation. This allows achieving large voltage swing out of the driver without any linearity constraint. However, for a linear-drive application, where the optical modulator is to be driven by an incoming PAM4 data-stream, none of these options are feasible.

A linear high-swing driver should have small output impedance, a fixed g_m -source and large enough supply voltage to support the high swing operation, such as the recently demonstrated complementary pseudo-differential cascode linear driver [24]. We utilize this driver as an example driver architecture to illustrate the electro-optic co-optimization framework. The driver schematic is shown in Fig. 8(a). The EO phase shifters in each arm of the MZM are realized by a PN-junction diode operating in reverse bias. The design choices for this

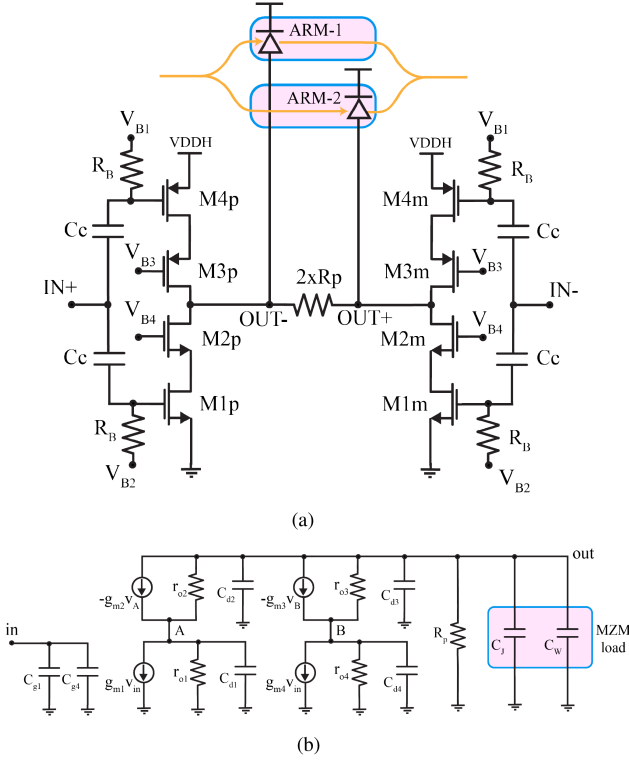


Fig. 8. MZM driver [24] used for the co-optimization framework: (a) schematic, (b) small-signal model.

driver topology can be understood as follows: i) large supply voltage is supported by cascoding the transistors, ii) getting rid of any tail current source by AC-coupling the signal to the input transistors helps in saving significant voltage head-room, iii) using complimentary topology reduces the power consumption by $2\times$ by means of sharing the bias current, iv) the shunt passive resistance R_p improves bandwidth, gives better control of the low-frequency gain and increases linearity range by reducing the low-frequency gain.

The small-signal model of the driver is shown in Fig. 8(b). Here, C_J is the effective junction capacitance of the PN-junction diode and C_W represents the wiring capacitance due to the metal routing from the driver output to the MZM. Both of them are linearly proportional to the length, L of the MZM, i.e., $C_J = C_{J0} \times L$ and $C_W = C_{W0} \times L$. Note that although this lumped capacitance approximation is accurate only for MZM lengths smaller than $500 \mu\text{m}$ (i.e., $1/10$ -th of the electrical wavelength), for longer MZM, the driver can be segmented where each driver segment is driving a specific section of the MZM resulting in a same value of net junction capacitance [17]. Now, assuming that the PMOS and NMOS devices are sized to have similar transconductance and intrinsic gain, i.e., $g_{m1} = g_{m4} = g_m$; $r_{o1} = r_{o4} = r_o$ and $g_{m2} = g_{m3} = g_{m,c}$; $r_{o2} = r_{o3} = r_{o,c}$, the mid-band gain and 3-dB bandwidth of the driver can be expressed as:

$$H_{dr,0} = \frac{2g_m R_p (r_{o1} + g_{m,c} r_{o1} r_{o2})}{r_{o1} + r_{o2} + R_p + g_{m,c} r_{o1} r_{o2}} \quad (27)$$

$$\omega_{3dB} = \frac{1}{(R_p || \frac{g_{m,c} r_{o1} r_{o2}}{2}) (C_J + C_W + C_{par})} \quad (28)$$

Here C_{par} is the sum of all the drain parasitic capacitance and can be given by $C_{par} = \gamma g_m / (\pi f_T)$, where γ is the drain-to-gate capacitance ratio and f_T is the transit frequency. In reality, the combination of junction and wiring capacitance is much higher than the parasitic drain capacitances of the devices. As in practice, $R_p \ll g_{m,c} r_{o1} r_{o2}$, the mid-band gain can be approximated as $H_{dr,0} \approx 2g_m \kappa R_p$, where $\kappa = (r_{o1} + g_{m,c} r_{o1} r_{o2}) / (r_{o1} + r_{o2} + g_{m,c} r_{o1} r_{o2})$. Note that the driver gain is directly proportional to R_p and the bandwidth is inversely proportional to R_p . Hence, g_m and R_p can be properly chosen to achieve a target gain and bandwidth combination.

Another important consideration for the driver is the linearity. The input linearity is dependent on the choice of overdrive voltage, V^* for the input transistors and the choice of the supply voltage or voltage headroom decides the output linearity. As a rule of thumb, for a given input peak-to-peak single-ended swing of $V_{TX,in}$, the minimum required overdrive voltage to maintain good linearity is given by, $V^* = V_{TX,in} / \sqrt{2}$. Similarly the minimum required supply voltage to maintain a good output linearity can be derived as,

$$V_{DDH,min} = 2V_1^* + 2V_2^* + H_{dr,0} * V_{TX,in} \quad (29a)$$

$$= V^* (4 + \sqrt{2} H_{dr,0}) \quad (29b)$$

Here it is assumed that all the devices have identical V^* . As an example, for an input peak-to-peak signal swing of 0.5 V and a low-frequency driver mid-band gain of 3-dB , the required V^* is $\sim 354 \text{ mV}$ and the minimum required supply voltage is $\sim 2.12 \text{ V}$. Considering this minimum required supply voltage, the power consumption and energy efficiency of the driver can be derived as

$$P_{drv} = V_{DDH} \times I_{Bias} \quad (30a)$$

$$= V^* (4 + \sqrt{2} H_{dr,0}) \times g_m V^* \quad (30b)$$

$$= (4 + 2\sqrt{2} g_m \kappa R_p) \times g_m V^{*2} \quad (30c)$$

$$E_{b,drv} = \frac{P_{drv}}{2f_{symbol}} \quad (31a)$$

$$= (2 + \sqrt{2} g_m \kappa R_p) \times g_m V^{*2} / f_{symbol} \quad (31b)$$

where f_{symbol} is the target symbol-rate of the electrical data, which is 56 GSym/s in this work.

V. BANDWIDTH-AWARE ELECTRO-OPTIC CO-OPTIMIZATION FRAMEWORK

The previous sections analyze and model the optical and electrical data-path of the coherent transmitter. The overall energy efficiency of the transmitter can be divided into the electrical and optical energy-efficiency. While the optical power consumption is governed only by the laser power, the electrical power is distributed into the driver, biasing circuits and thermal phase shifters. Power consumed in the biasing circuits is often comparatively smaller. On the other hand, the thermal phase-shifters

TABLE I
OPTICAL PARAMETERS

Parameter	Description	Value
α_p	Phase-shifter power absorption coefficient	8 dB/cm
α_f	Phase-shifter field absorption coefficient	$\alpha_p/2$
α_s	Splitter / combiner insertion loss	-0.1 dB
$V_\pi L_\pi$	Phase-shifter modulation efficiency	16 V.mm
ω_{EO}	MZM intrinsic EO bandwidth	32 GHz
α_c	Edge-coupler insertion loss	-2 dB
α_r	LO phase-tuner insertion loss	-0.1 dB
α_h	Optical-hybrid insertion loss	-2 dB
$\mathcal{R}$	Photo-diode responsivity	0.9 A/W
η_w	Laser wall-plug efficiency	10%
η_m	Link margin	3 dB

TABLE II
ELECTRICAL PARAMETERS

Parameter	Description	Value
f_{symbol}	Symbol-rate of the electrical PAM4 data	56 GSym/s
$V_{TX,in}$	Transmitter input swing (differential peak)	0.5 V
V^*	Transistor overdrive voltage	$\frac{V_{TX,in}}{\sqrt{2}}$
f_T	Technology transit frequency	210 GHz
γ	Drain-to-gate capacitance ratio	1
A_0	Intrinsic gain of input transistors	5
g_m	Transconductance of the input transistors	5-100 mS
r_o	Output impedance of the input transistors	A_0/g_m
R_p	Shunt resistance added to the driver	5-200 Ω
$g_{m,c}$	Transconductance of the cascode transistors	g_m
$r_{o,c}$	Output impedance of the input transistors	r_o
C_{J0}	Unit junction capacitance of MZM	0.25 pF/mm
C_J	Junction capacitance of MZM	$C_{J0} \times L$
C_{W0}	MZM unit routing capacitance	0.2 pF/mm
C_W	MZM routing capacitance	$C_{W0} \times L$
$\bar{i}_{nT}$	TIA input referred noise	2 μ A

consume static electrical power which is independent of the data rate or any link design parameters. Hence, to optimize the overall energy efficiency of the transmitter, one needs to minimize the sum of the driver and laser power. In this section we describe this electronic-photonics co-optimization framework taking into account the limited bandwidth of the transmitter. Tables I and II list the optical and electrical parameters used in this co-optimization example.

A. Optimizing Transmitter EO Bandwidth

Considering (8, 10, 20), the frequency domain transfer function of the coherent transmitter from the MZM-driver input in the TX to the balanced-PD output in the RX can be found out as,

$$H_{TX}(s) = h_{opt} \sqrt{\frac{k(1-k)}{2}} P_{laser} e^{-\alpha_f L} H_{dr}(s) H_{ps}(s) \quad (32)$$

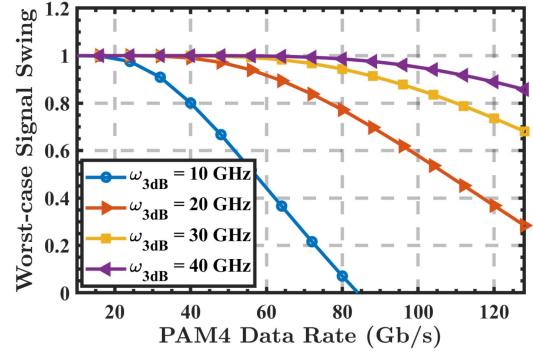


Fig. 9. Effect of limited transmitter bandwidth: normalized worst-case signal swing vs PAM4 data rate. The signal swing degrades for higher data rate due to insufficient settling time and inter-symbol interference [30].

where $h_{opt} = \mathcal{R} \alpha_h \alpha_c^2 \alpha_s^2 \sqrt{\alpha_r} / 2$ is a constant dependent on several optical parameters. Equation (32) indicates that $H_{TX}(s)$ depends on several design parameters namely the laser splitting ratio (k), laser power (P_{laser}), MZM length (L), driver g_m and R_p . A solution in this 5-dimensional design space is a feasible set of design parameters which satisfies the BER criterion at the balanced-PD output. Note that, as we found in Section III, the SNR monotonically increases with increase in the laser power. This implies, for any set of other design parameters, one can always find a minimum value of P_{laser} to meet the target BER. Therefore, P_{laser} is used as a dependent variable in this co-optimization framework.

Form (32), it is evident that the frequency behavior of the overall transmitter is jointly governed by the frequency behavior of the driver and the MZM EO phase-shifter. The EO phase-shifter bandwidth is ultimately constrained by the device design and the manufacturing process parameters. Therefore, even if one tries to push the driver bandwidth higher, the overall transmitter bandwidth will be limited by the MZM. Hence, it is crucial to incorporate the bandwidth constraint into the co-optimization framework. Note that the mid-band value of $H_{TX}(s)$ determines the received peak-to-peak signal amplitude as given by (21b). At much higher data rates, (21b) just indicates the envelope of the received signal. However, as shown in Fig. 9, the worst-case signal swing and also the corresponding SNR degrades with an increase in the data rate due to insufficient settling time and inter-symbol interference (ISI) caused by limited transmitter bandwidth [30], [31]. A useful metric to quantify this worst-case signal swing is the vertical eye-opening (VEO) [30].

Using the transfer function in (32), one can find the current pulse response at the balanced-PD output numerically. The VEO of the inner PAM4-eye can be obtained from the current pulse-response as,

$$VEO = \frac{h_0}{3} - \sum_{i \neq 0} |h_i| \quad (33)$$

where h_0 is the main cursor and $h_i (i \neq 0)$ are pre and post-cursor ISI terms of the pulse-response. The factor of 1/3 accounts for the fact that the signal has 4 amplitude levels. Thus, considering the effect of bandwidth limitation, the BER expression given by

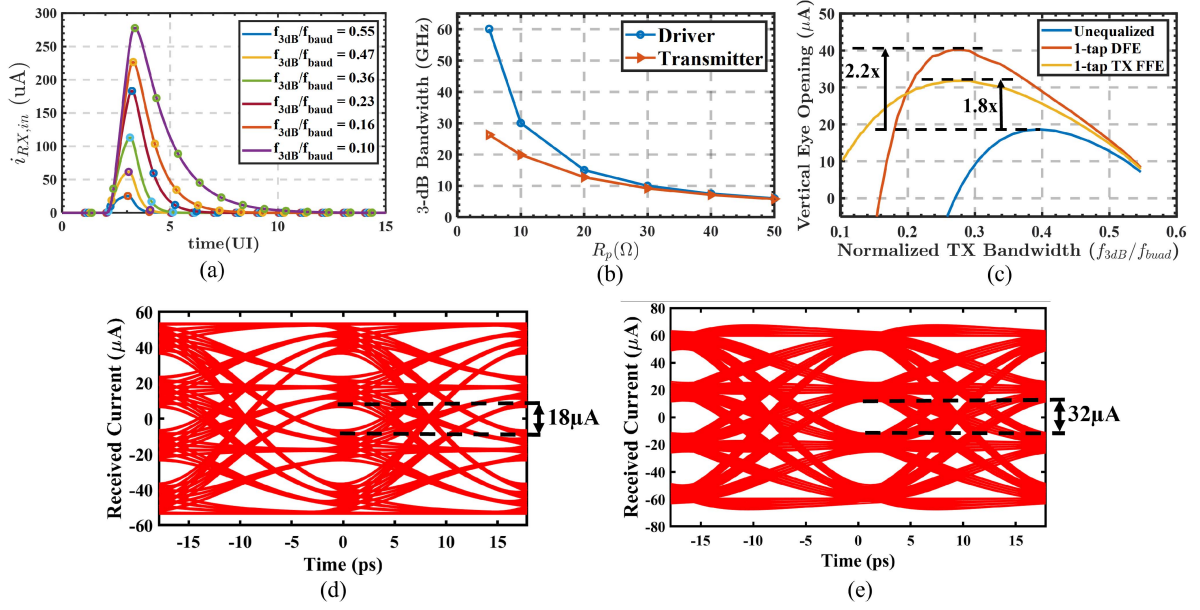


Fig. 10. Transmitter bandwidth optimization: (a) current pulse-response for different TX 3-dB bandwidths normalized by the symbol-rate (f_{symbol}), (b) dependence of driver and TX 3-dB bandwidth on R_p , (c) variation of vertical eye-opening (VEO) with normalized TX bandwidth for different equalization scenarios, (d) simulated 112Gb/s current eye-diagram corresponding to the maximum VEO point of unequalized link and (e) equalized link with 1-tap TX FFE. For this analysis, the TX input differential swing is assumed to be 1 Vppd and k , P_{laser} , L and g_m are kept fixed to 0.5, 20mW, 1mm and 50mS respectively.

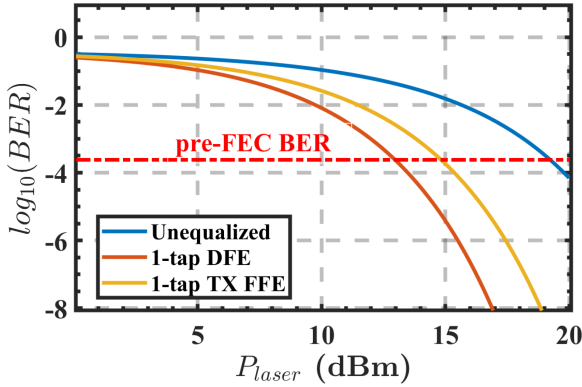


Fig. 11. Change of bit-error rate (BER) with total laser power. For each value of P_{laser} , the BER corresponds to the maximum VEO. k , L and g_m are kept fixed to 0.5, 1mm and 50mS respectively.

(26b) can be modified as

$$BER = \frac{3}{8} \operatorname{erfc} \left(\frac{VEO}{2\sqrt{2} \times \sqrt{i_{ns}^2 + i_{nT}^2}} \right) \quad (34)$$

Fig. 10(a) shows the current pulse response for different TX 3-dB bandwidths. In this analysis, R_p is swept to achieve different driver bandwidths, while k , P_{laser} , L and g_m are kept fixed to 0.5, 20 mW, 1 mm and 50 mS respectively. Larger values of R_p result in large driver mid-band gains and low driver bandwidths which generate larger current pulses with significant ISI. Conversely, smaller values of R_p result in small driver mid-band gains and high driver bandwidths which yield small current pulses with negligible ISI.

Fig. 10(b) shows the relationship of TX 3-dB bandwidth and the driver 3-dB bandwidth vs R_p . These plots indicate that as the driver bandwidth increases, the TX bandwidth becomes limited

by the EO phase-shifter. Fig. 10(c) shows the dependence of the VEO on the TX 3-dB bandwidth for the unequalized case and also with a 1-tap decision-feedback equalizer (DFE) and a 1-tap TX FFE deployed in the host-SerDes. For the last case, peak power constraint of the host-SerDes TX is addressed. For a smaller TX bandwidth, ISI and insufficient settling time dominate closing the inner-eye (i.e., $VEO < 0$). For larger TX bandwidth, gain becomes very small degrading the signal amplitude. Hence, best VEO can be achieved at a moderate bandwidth which is $\sim 0.4f_{symbol}$ for unequalized link and $0.28f_{symbol}$ for a link equalized by either a 1-tap DFE or a 1-tap TX FFE. Note that, as the shot noise at the TIA input (i_{ns}) is dominated by the LO power which is kept fixed, the SNR is directly proportional to the VEO in this analysis. As Fig. 10(c) indicates, 1-tap TX FFE gives $1.8\times$ VEO improvement compared to the unequalized case and a $2.2\times$ improvement in the VEO can be achieved with a 1-tap DFE. These correspond to 5.1 dB and 6.8 dB SNR improvements respectively. The same analysis can be done for more complex combination of equalization schemes to see the VEO improvements. However, this work deals with only 1-tap DFE and a 1-tap TX FFE for proof-of-concept. Fig. 10(d), (e) show the simulated 112Gbps eye-diagram using a PRBS13 sequence. The first eye-diagram corresponds to the maximum VEO point of the unequalized link and the later one corresponds to the maximum VEO point with the 1-tap TX FFE showing close match of the inner eye-opening with the VEO analysis in Fig. 10(c).

B. Laser Power Sensitivity

As discussed earlier, the SNR monotonically increases with increase in the laser power. Hence, after finding the optimum transmitter bandwidth that maximizes the VEO as well as the

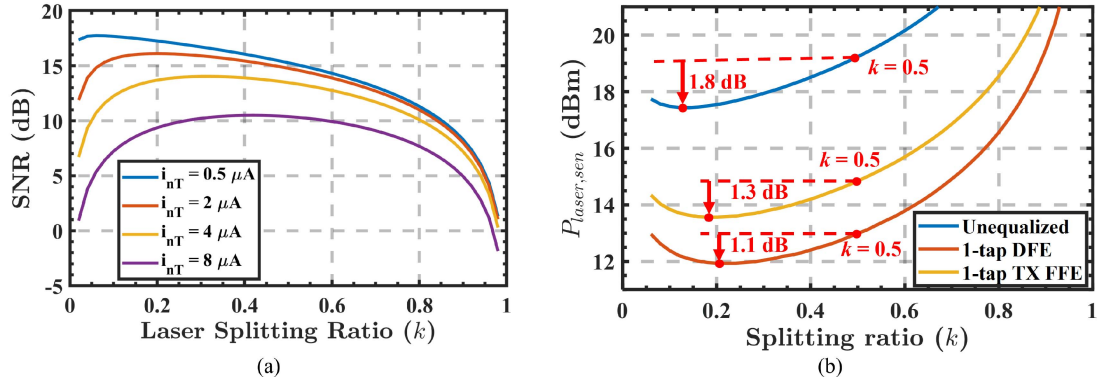


Fig. 12. Optimizing the laser splitting ratio: (a) variation of SNR for different RX-AFE thermal noise current (assuming a laser power of 13 dBm and 1-tap DFE equalization), (b) laser power sensitivity vs laser splitting ratio for different equalization scenarios. The later assumes an rms thermal noise current of 2 μA .

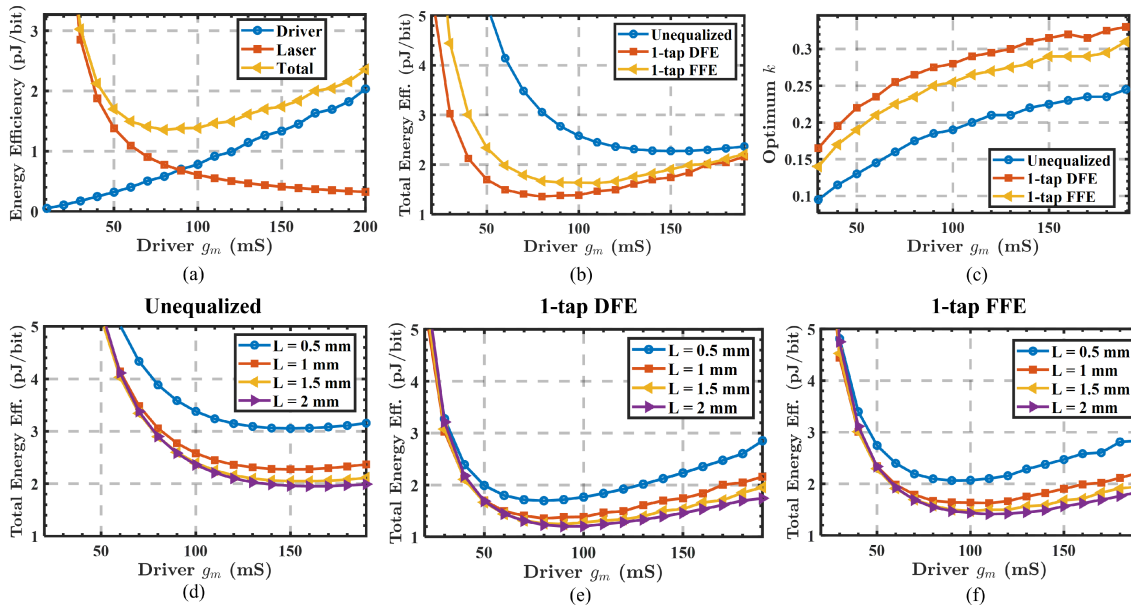


Fig. 13. Variation of driver and laser energy-efficiency with driver g_m : (a) driver, laser and total TX energy efficiency variation assuming a 1-tap DFE equalized link and 1 mm long MZM, (b) comparison of total driver + laser energy-efficiency for different equalization scenarios, (c) corresponding choices of optimum laser splitting ratio, (d-f) dependence of total energy-efficiency on MZM length for different equalization scenarios.

SNR, in the next step we find the laser power sensitivity, $P_{laser,sen}$, defined as the minimum required total laser power corresponding to the maximum VEO that meets the target BER. Assuming a KP-4 FEC, the target BER is assumed to be 2.4×10^{-4} . Fig. 11 shows the change in BER with the laser power for the same fixed k , L and g_m values as mentioned earlier. The $P_{laser,sen}$ for un-equalized link is 19.3 dBm whereas the links with 1-tap DFE and 1-tap FFE show 12.8 dBm and 14.8 dBm of laser power sensitivities respectively. For this analysis, the TIA input referred thermal noise current is assumed to be 2 μA as listed in Table II.

C. Optimizing the Laser Splitting-Ratio

As a next step, the splitting ratio, k is optimized. As already discussed in Section (III.B.2), the optimum splitting ratio should maximize the SNR, therefore minimizing the laser power

sensitivity value. From (26a) it can be intuitively understood that when the total noise is dominated by TIA thermal noise, the best possible BER and SNR can be achieved for $k = 0.5$, as it maximizes the signal swing. However, when shot-noise becomes comparable to the thermal noise or when the total noise is dominated by the shot-noise, the optimum point changes due to the k -dependency of shot-noise.

Fig. 12(a) shows the variation of SNR with splitting ratio for different values of TIA input-referred thermal noise. Here, the link is assumed to utilize 1-tap DFE and P_{laser} , L , g_m are kept fixed to 20 mW (=13 dBm), 1 mm and 50 mS respectively. If thermal noise current is assumed to be large enough, the maximum SNR is obtained near $k = 0.5$. However, for most of the state-of-the-art linear-TIA designs, the input-referred rms noise current value varies between 1–4 μA [32], [33], [34], [35] and for monolithic processes, it can be even smaller [15], [36]. For these realistic values of thermal noise, the maximum

SNR-point shifts to lower k values as Fig. 12(a) indicates. For example, for thermal noise currents of $4 \mu\text{A}$, $2 \mu\text{A}$ and $0.5 \mu\text{A}$ the optimum splitting ratio values turn out to be 0.32, 0.2 and 0.08 respectively. Going forward, this co-optimization framework assumes a thermal noise current of $2 \mu\text{A}$.

Fig. 12(b) plots the laser power sensitivity for different laser splitting ratios, assuming $L = 1 \text{ mm}$ and $g_m = 50 \text{ mS}$. For the un-equalized link, a 1.8 dB improvement in the laser power sensitivity can be achieved by choosing the optimum k -value as compared to $k = 0.5$. Similarly, links equalized with either a 1-tap DFE or 1-tap FFE shows an improvement of 1.1 dB and 1.3 dB respectively. As the laser energy efficiency is directly proportional to the laser power sensitivity (23a), optimizing the laser splitting ratio directly improves the overall energy efficiency of the link.

D. Optimum Energy-Efficiency

So far the co-optimization scheme assumes a fixed MZM length and driver g_m while optimizing the laser energy efficiency. As a next step, different driver g_m values can be chosen to see its impact on the laser energy efficiency and the total transmitter energy efficiency.

Fig. 13(a) plots the driver, laser and total TX energy efficiency values for different choices of driver g_m assuming a 1-tap DFE equalized link and 1 mm long MZM. For each driver g_m value, the laser splitting ratio is optimized and the corresponding laser power sensitivity is used to calculate the laser energy efficiency. Similarly, the driver energy efficiency has been found using (31). Note that for small values of g_m , the link needs relatively larger amount of laser power and the overall energy-efficiency is dominated by the laser power. The laser energy-efficiency improves as the driver becomes stronger (i.e., higher g_m) and for sufficiently large g_m , the driver power dominates the overall energy-efficiency. The optimum value of the total energy efficiency is found to be 1.35 pJ/bit which corresponds to $g_m = 80 \text{ mS}$. The maximum laser power utilized by a single link is often constrained by the laser source capacity. Similarly, in certain scenarios with an external laser source (ELS), the electrical energy-efficiency is more constrained due to the electrical power budget and package thermal considerations. Hence, this analysis also helps in figuring out the optimal design choices for either a given driver power constraint or a given laser power constraint.

Fig. 13(b) compares the total energy-efficiency for all 3 different equalization scenarios. The unequalized link shows an optimum TX energy-efficiency of 2.27 pJ/bit which improves down to 1.35 pJ/bit with a 1-tap DFE and 1.63 pJ/bit with a 1-tap FFE. Hence, any equalization scheme in the host SerDes can significantly improve the energy-efficiency of the coherent TX. The corresponding optimum laser splitting ratio, k , is shown in Fig. 13(c). The variation of optimum k can be intuitively understood as follows. As g_m increases, the required laser power decreases which reduces the shot noise contribution at the balanced-PD output. This in turn increases the optimum k values as evident from Fig. 25(a). Similarly, with equalization, the required laser power reduces making the optimum k higher compared to the unequalized case. Finally, Fig. 13(d)–(f) captures the effect of MZM length on the total TX energy

Algorithm 1: Transmitter EO Co-optimization Routine.

```

1: Assign optical parameter values
2: for  $L$  in the MZM design range do
3:   Calculate  $C_J$  and  $C_W$ 
4:   Define  $H_{ps}(s)$ 
5:   for  $g_m$  in [5-200] mS do
6:     for  $s$  in [0-1] do
7:       for  $P_{laser}$  in [1-100] mW do
8:         for  $R_p$  in [5-100]  $\Omega$  do
9:           Define  $H_{dr}(s)$ 
10:          Define  $H_{TX}(s)$ 
11:          Find the current pulse response
12:          Apply equalization (optional)
13:          Find the vertical eye-opening ( $VEO$ )
14:          if  $VEO$  reaches maxima then
15:             $VEO_{max} \leftarrow VEO$ 
16:             $R_{p,opt} \leftarrow R_p$ 
17:            break
18:          end if
19:        end for
20:        Calculate the shot-noise and total noise
21:        Calculate the bit-error-rate (BER)
22:        if BER reaches target value then
23:           $P_{laser,sen} \leftarrow P_{laser}$ 
24:          break
25:        end if
26:      end for
27:      if  $P_{laser,sen}$  reaches minima then
28:         $s_{opt} \leftarrow s$ 
29:        store all parameters
30:        break
31:      end if
32:    end for
33:    Calculate  $E_{b,dr}$ ,  $E_{b,laser}$  and  $E_{b,total}$ 
34:  end for
35:  Find  $g_{m,opt}$  which minimizes  $E_{b,total}$ 
36: end for

```

efficiency. The allowable MZM length depends largely on the die real-estate which makes accommodating very long MZM difficult in state-of-the art silicon-photonics processes. The analysis in this work suggests that although the achievable energy-efficiency initially improves with increase in MZM length, the improvement saturates after a certain length ($\sim 1 \text{ mm}$) due to the optical loss, bandwidth and phase-shift trade-off. Hence, depending on the silicon area and overall power budget one can choose a certain MZM length following the analysis in this paper. The entire co-optimization routine is summarized in Algorithm 1.

VI. CONCLUSION

This paper proposes an electronic-photonics co-optimization framework based on transmitter pulse-response to optimize the overall energy-efficiency of O-band coherent transmitters. The framework guides the designer to choose optimum optical and electrical design parameters for some given constraints on

electrical and laser power budgets, while taking into account the bandwidth limitations and link equalization capabilities. The comprehensive modelling of optical data path starting from a single optical component enables the designer to model more complicated optical transceivers involving other types of optical devices.

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